FestPass Jenkins Assignment: Evidence

Cloud and Jenkins | Student Laptop Lab | 30 October 2026

Student alias	Prisha (from Git commit author)
Class code	001
Challenge code	00376
Jenkins job	festpass-ci (Jenkins 2.580.1)
Agent	student-laptop

Summary

	Red build	Green build
Build number	#2	#3
Commit	8a00c6a "Personalize my help introduction"	35953c1 "Fix cancel: remove student so the seat is freed"
Cause	Started by an SCM change	Started by an SCM change
Result	**FAILURE**	**SUCCESS**
Tests	28 tests, 3 failures	28 tests, 0 failures
Product stages	Build JAR, Smoke test JAR, Save artifact skipped	All stages ran
JAR	Not published	festpass.jar published (SHA-256 d9f72674cc...)

1. Job configuration and polling trigger

- Pipeline job **festpass-ci**, Pipeline script from SCM, Git repository `git://127.0.0.1:9418/festpass-java`, no credentials, branch `*/main`, script path `Jenkinsfile`.
- Polling trigger committed in the Jenkinsfile: `triggers { pollSCM('* * * * *') }`, which checks Git once per minute. No duplicate UI trigger and no Build periodically.
- Build #1 was the single manual bootstrap run that loaded the trigger. Builds #2 and #3 started automatically.



Figure 1. Job page: builds #1 and #2 failed, #3 succeeded. The test trend moves from 3 failures to all 28 passing.

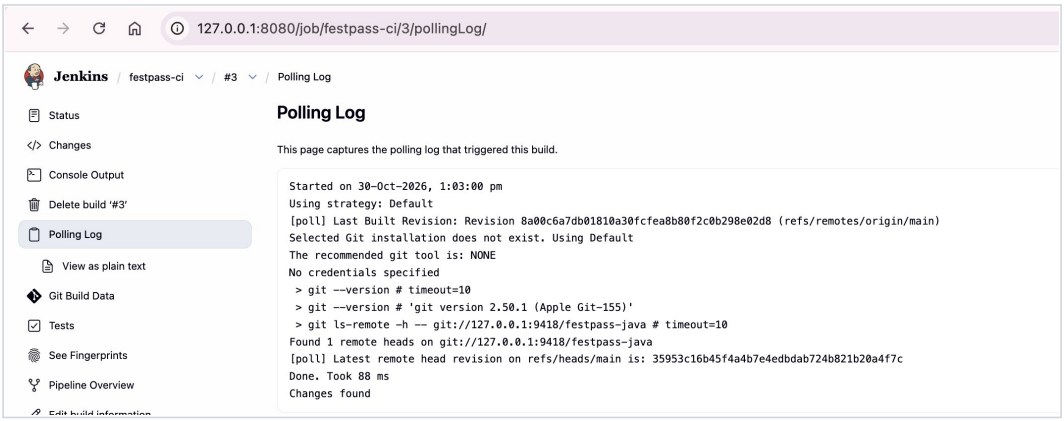


Figure 2. Polling log for build #3: Jenkins compared the last built revision (8a00c6a) with the new head (35953c1), found changes, and started the build.

2. Automatic red build (#2)

- After committing the personalised help message (commit **8a00c6a**), I did not click Build Now. Polling started build #2 automatically.
- The build failed at the Test stage with 3 of 28 tests failing. Checkout, Submission setup and compilation succeeded.
- Build JAR, Smoke test JAR and Save artifact were skipped, so no JAR was published. Only the diagnostic submission-evidence.csv was archived.

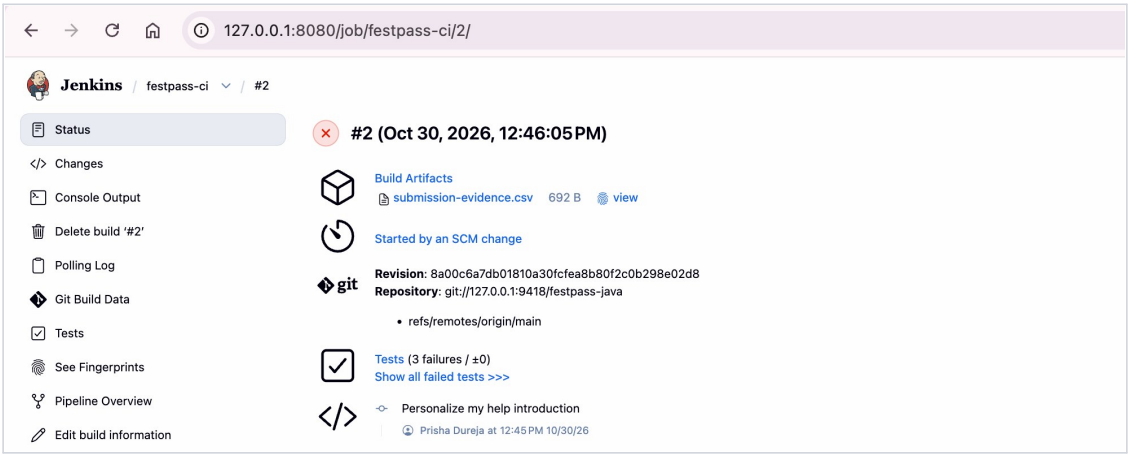


Figure 3. Build #2: Started by an SCM change, revision 8a00c6a, 3 test failures, only submission-evidence.csv archived.

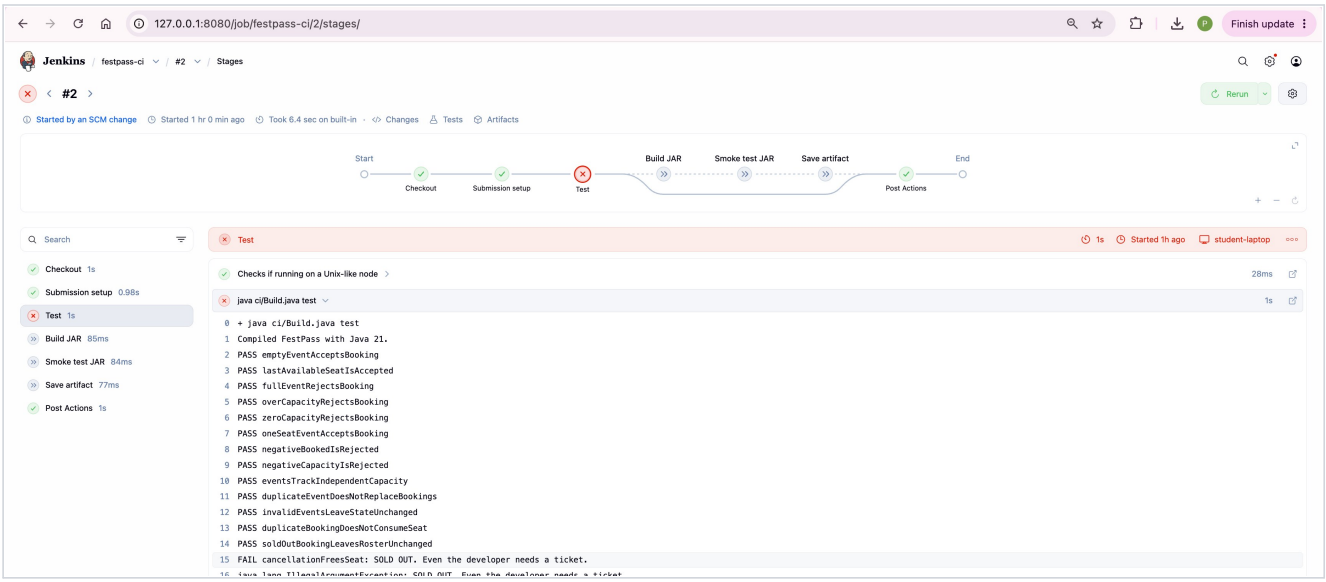


Figure 4. Build #2 stages: Test failed; Build JAR, Smoke test JAR and Save artifact were skipped. The stage ran on agent student-laptop.

3. Failing assertion and diagnosis

Failing test	FestPassTest.cancellationFreesSeat
Error message	`java.lang.IllegalArgumentException: SOLD OUT. Even the developer needs a ticket.`
Other failures	cancellationsSurviveReload ("Cancellation was lost") and cliWorkflowPersistsAcrossCommands ("List does not reflect persisted cancellation")

Business rule violated: when a student cancels a booking, their seat must be released so that another student can book it. The test fills an event, cancels one booking and then books a new student. The new booking was rejected as SOLD OUT because `cancel()` in BookingService.java only checked that the booking existed and never removed the student, so the event still counted as full. The same missing removal explains the other two failures: the cancellation was never saved, so it was lost after reloading the data file.

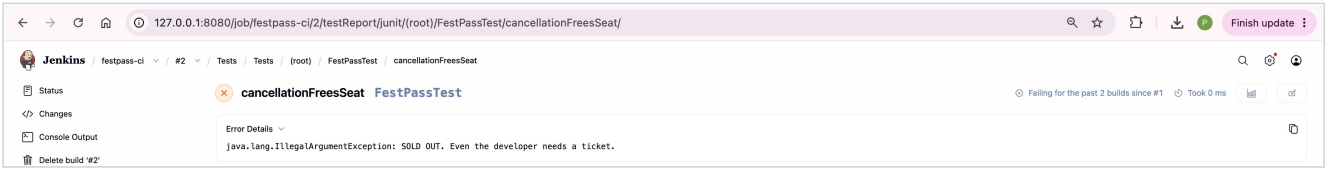


Figure 5. Jenkins test report for cancellationFreesSeat in build #2.

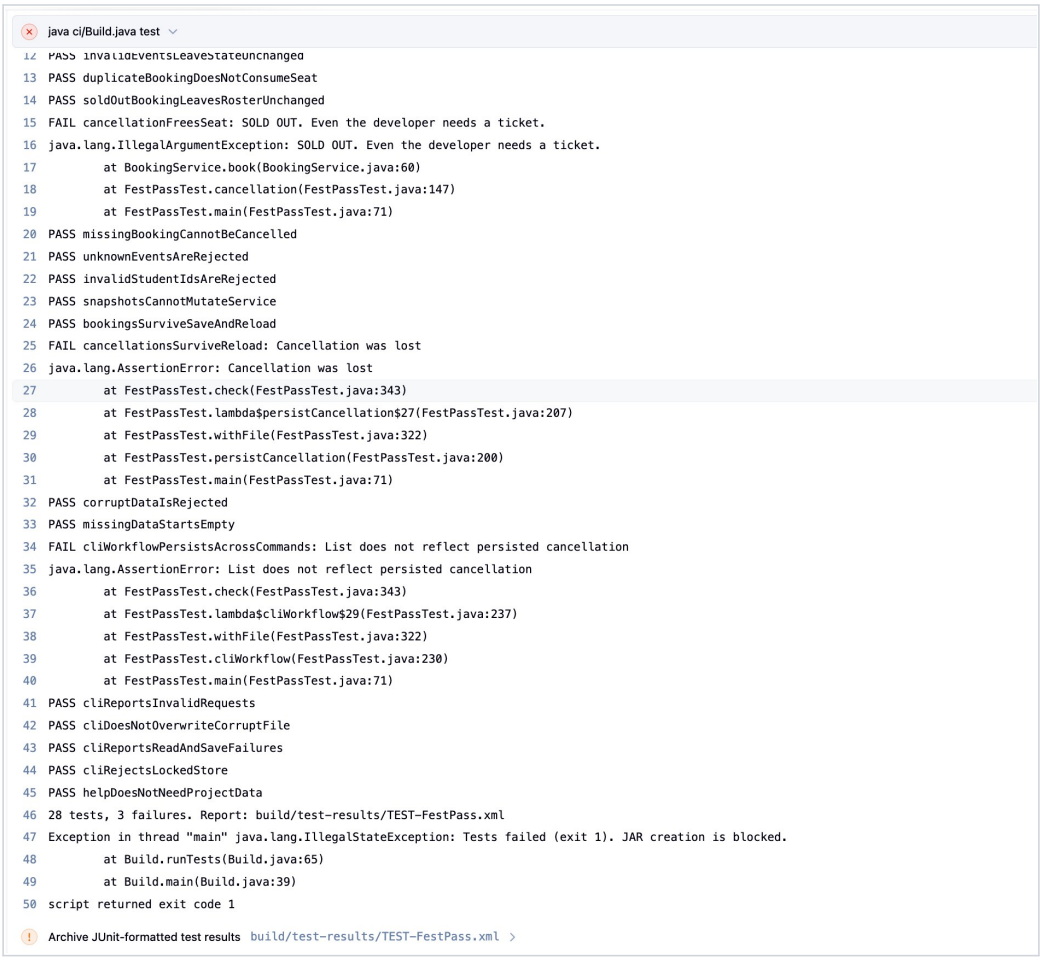


Figure 6. Build #2 test log: 28 tests, 3 failures, and "JAR creation is blocked".

4. The fix

I added one line to `cancel()` in `src/main/java/BookingService.java`, after the check that the booking exists:

```
    public void cancel(String eventId, String studentId) {
        validateId(studentId);
        Registration event = requireEvent(eventId);
        if (!event.students.contains(studentId)) {
            throw new IllegalArgumentException("No booking for student: " + studentId);
        }
+       event.students.remove(studentId);
    }
```

- No tests, build helpers, evidence code or pipeline stages were changed. Both evidence CSVs record the same TestSourceSha256 (d203243b...) and PipelineSha256 (8fb5de75...).
- Before committing, I verified locally:

```
$ java ci/Build.java verify
Compiled FestPass with Java 21.
...
28 tests, 0 failures. Report: build/test-results/TEST-FestPass.xml
BUILD SUCCESS: build/festpass.jar

$ java ci/SmokeTest.java
PASS: seven real-JAR commands; persistence, sold-out rejection,
      cancellation, and seat reuse verified.

$ git commit -m "Fix cancel: remove student so the seat is freed"
[main 35953c1] Fix cancel: remove student so the seat is freed
 1 file changed, 1 insertion(+)
```

Note: my Mac's shell environment caused a Java zip-library error locally, so I ran the local verify, smoke test and JAR in a clean environment (env -i ... with JDK 21). The Jenkins builds were unaffected.

5. Automatic green build (#3)

- After committing the fix (commit **35953c1**), I did not click Build Now. Polling started build #3 automatically and it succeeded with 28 tests and 0 failures.
- Artifacts: festpass.jar, build-info.txt, smoke-test.txt and submission-evidence.csv.

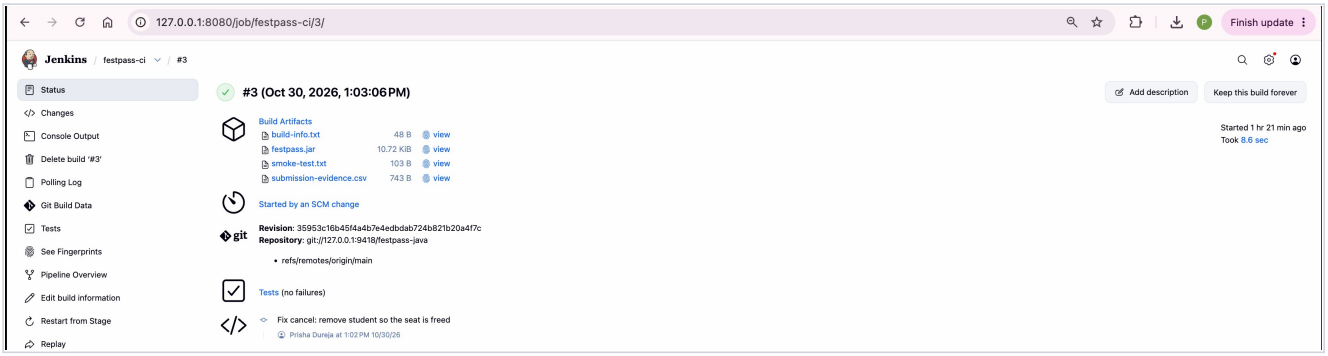


Figure 7. Build #3: Started by an SCM change, revision 35953c1, no test failures, all four artifacts.

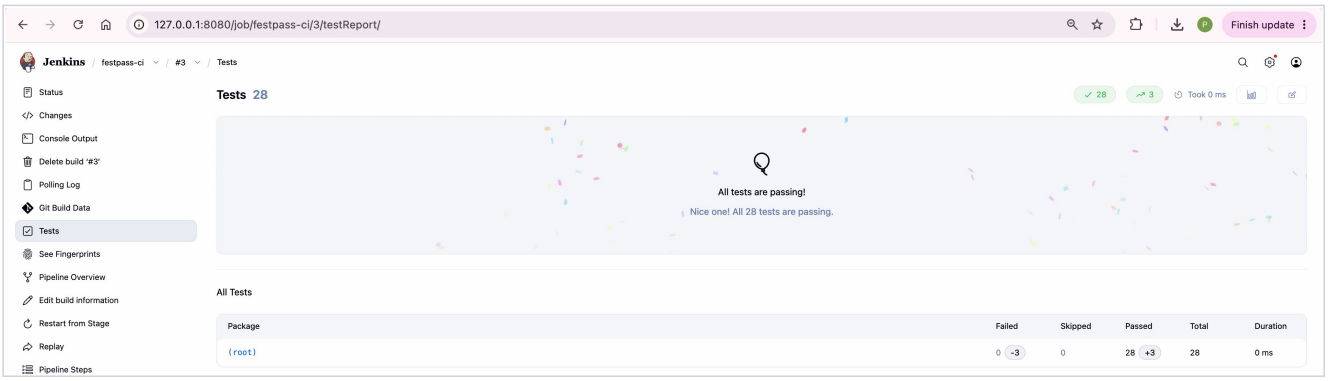


Figure 8. Build #3 test report: all 28 tests passing (3 fewer failures than build #2).

6. Final JAR

I downloaded festpass.jar and submission-evidence.csv from build #3 into one folder, and the build #2 CSV into a separate folder. Running the downloaded JAR:

```
$ java -jar festpass.jar help
FestPass - campus event registration (by Prisha)
Usage: java -jar build/festpass.jar [--data FILE] COMMAND
  add-event ID "TITLE" CAPACITY
  list
  book EVENT_ID STUDENT_ID
  cancel EVENT_ID STUDENT_ID
  roster EVENT_ID
  summary
  help
...

$ shasum -a 256 festpass.jar
d9f72674ccea88e72e69156731da52d2b08ad910499e47751704e3d10dc9b593  festpass.jar
```

The hash matches the JarSha256 recorded in the build #3 evidence CSV, so this is exactly the JAR Jenkins built and tested.

7. Evidence CSVs

Field	Red: build #2	Green: build #3
ClassCode / ChallengeCode	001 / 00376	001 / 00376
StudentAlias	Prisha (git-commit-author)	Prisha (git-commit-author)
Commit	8a00c6a7db01...	35953c16b45f...
RecordedAtUtc	2026-10-30 07:16:11	2026-10-30 07:33:14
Result	FAILURE	SUCCESS
Tests / Failures	28 / 3	28 / 0
JarSha256	NOT_PUBLISHED	d9f72674ccea...
TestSourceSha256	d203243b893e...	d203243b893e... (unchanged)
PipelineSha256	8fb5de757d33...	8fb5de757d33... (unchanged)

Submitted files: festpass-java source repository, red-build-2/submission-evidence.csv, green-build-3/submission-evidence.csv, green-build-3/festpass.jar, and this evidence document.